Advait Paranjpe

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Education

University of California, Los Angeles | *M.S. in Electrical and Computer Engineering* Sep 2025 – Dec 2026

- Coursework: CPU/GPU Architecture, Memory Systems, Cache Coherence, Interconnect Design.
- GPA: 3.7/4.0

The University of Manchester | *B.Eng. in Electrical and Electronic Engineering* Sep 2022 – Jul 2025

- Dissertation: "Cascaded PLL Microwave Synthesizer for 5G systems."
- **First Class Honours** – highest UK degree classification.

Experience

Jaguar Land Rover North America | *Incoming Semiconductor Applications Engineering Intern* Jul 2026 – Sep 2026

VAST Lab, UCLA | *Student Researcher – GPU/FPGA Acceleration* Jan 2026 – Mar 2026

- Profiled ROCm attention kernels and memory bottlenecks to identify FPGA acceleration opportunities for Engram LLM inference.
- Modeled FPGA cache architectures, increasing projected hit rate from ∼5% at 256 entries to ∼48% with a 64k-entry, 4-way design.
- Translated workload profiling into FPGA tradeoffs across cache capacity, lookup throughput, locality, and on-chip memory usage.

NESO / National Grid ESO | *Software Engineering Intern (two summers)* Jul 2024 – Sep 2025

- Re-architected a 100k+ LOC simulation engine through vectorization and pipeline optimization, cutting runtime by 10x.
- Optimized 100M+ records through chunked, memory-efficient execution, cutting RAM from 32 GB to 16 GB and runtime by 11x.
- Reduced CPU context-switching overhead through parallel API batching and staged processing, increasing throughput by 24x for time-sensitive market telemetry.

Projects

Dementia Speech Classification | *Multimodal ML Pipeline (Python, PyTorch)* Mar 2026 – Jun 2026

- Built a multimodal pipeline combining TF-IDF linguistic features, HuBERT acoustic embeddings, and structured LLM-derived semantic features.
- Improved held-out performance to 0.926 AUROC, 0.854 F1, and 0.879 accuracy using calibrated late fusion, with 0.963 AUROC on a blind evaluation set.
- Developed reproducible feature extraction, model comparison, threshold analysis, error analysis, and blind-evaluation workflows.

SparrowML | *Hardware-Aware Edge ML Compiler and Runtime (Python, C, PyTorch)* May 2026 – Jun 2026

- Built a WISDM activity-recognition pipeline spanning PyTorch training, INT8 quantization, tensor serialization, C runtime generation, and RTL execution.
- Achieved 0.920 INT8 macro-F1 with a 0.009 drop from FP32 and 98.7% agreement with floating-point predictions.
- Benchmarked scalar, dense-vector, 2:4 sparse-vector, and multicore targets across accuracy, cycles, memory traffic, operations skipped, and utilization.

SpeakUp | *Quantized Edge Vision System (Python, TensorFlow Lite, C++)* Oct 2024 – Apr 2025

- Fine-tuned and quantized a MobileNetV2-based visual recognition model for real-time deployment on constrained edge hardware.
- Achieved approximately 91% recognition accuracy, 18 FPS throughput, and 180 ms end-to-end latency through model and pipeline optimization.
- Built the capture, preprocessing, inference, and output pipeline while profiling CPU, memory, latency, and power constraints.

Skills

- **Languages:** Python, C++, C, SQL, Bash, MATLAB, JavaScript, SystemVerilog.
- **Machine Learning:** PyTorch, TensorFlow Lite, scikit-learn, NumPy, Pandas, quantization, feature engineering, model evaluation, NLP, speech ML, and computer vision.
- **ML Systems:** inference profiling, hardware-aware optimization, data pipelines, HIP/ROCm, Docker, Git, Linux, and reproducible experimentation.